

# Patents as Indicators of Technological Change and Innovation — An Historical Analysis of the Patent Data 1830–1914

Ian INKSTER

To cite this article: Ian INKSTER (2003) Patents as Indicators of Technological Change and Innovation — An Historical Analysis of the Patent Data 1830–1914, Transactions of the Newcomen Society, 73:2, 179-208, DOI: [10.1179/tns.2003.010](https://doi.org/10.1179/tns.2003.010)

To link to this article: <https://doi.org/10.1179/tns.2003.010>



Published online: 31 Jan 2014.



Submit your article to this journal [↗](#)



Article views: 55



View related articles [↗](#)



Citing articles: 2 View citing articles [↗](#)

---

# Patents as Indicators of Technological Change and Innovation — An Historical Analysis of the Patent Data 1830–1914

by

Ian INKSTER

*Read at the Science Museum, London, 13 March 2002*

## INTRODUCTION

The paper considers a portion of the data gathered under a research project on British and international patenting in the years approximately 1830–1914.<sup>1</sup> Clearly enough, the potentials of this subject are great and the possible avenues of enquiry are labyrinthine. Many historians of engineering and technology are familiar enough with some of the sources — all researchers at times investigate particular patents in their field, in unearthing biography, in identifying incremental improvements, breakthroughs or trends, or in wrestling with the intricacies of differing patent regulations. Whilst much of this is grist to the mill, the paper isolates just four themes that might be at once novel and of particular interest to historians of technology.

In particular, it is suggested that this sort of reasonably systematic data might force reconsideration of the value of the enormous amount of material now lying in patent offices, patent libraries and in the archives of individual patent attorneys. It is unnecessary to believe that every patent was worthwhile, nor that all patents were ever equal — many were outrageous, others were epochal. But much the same might be said, for instance, of all capital investments. The latter fact has never stopped economists or economic historians from counting rates of investment or from examining broad trends in capital formation. The four themes here isolated are those of; Patents as Measures of Something Significant; Patents and Sites of Inventive Activity; Patents as Guides to Inventors and their Worlds; and Patents as Things that may be Valued (as well as Counted). These themes are complementary to other work resulting from this and earlier research projects.<sup>2</sup>

## PATENTS AS MEASURES OF SOMETHING SIGNIFICANT

A most casual dismissal of patents *en masse* is that they do not accurately measure or differentiate adequately between invention, improvement and innovation — this as part of a bad old adage that runs ‘you can’t count inventions’. There is of course a sense in which that is true and mundane, for patented inventions differed enormously across time and place, and

almost as greatly within any one site of invention. So we might briefly consider that wonderful Black Country city described by Arthur Young in 1791 — and by many others since — as ‘the first manufacturing town in the world’. Even in the 1830–50 period Birmingham was predominantly a place of un-mechanised workshops, whose toy and gun manufactures, brass and ironmongery, remained dominated by craftsmen benefiting from the fundamental Black Country technological advances of an earlier era — mineral fuel, puddling and rolling, and steam power.<sup>3</sup> The Birmingham patents of the considered twenty-year period included 28% in metalworking, 14% in clothing and accessories, 6% in motive power, 5% in general machinery improvements, but 43% in a range of small product improvements covering most of the small business industry of the city, such as furniture and decorations, printed and embossed goods, heating and cooking appliances, gas burners and lamps, plate and window glass, and, of course, firearms. Such patents were mainly produced by skilled tradesmen. A major support mechanism was through patent partnerships — 12% of over 400 patents were through partnerships, but this represented 24% of all Birmingham patentees, most of whom were small manufacturers, millwrights, machinists and skilled tradesmen engaged in a self-help industrial culture.

It is certainly possible to show that patent trends are good reflections or measures of salient processes or events. Thus a civil war in patents may be found. With micro-investigation, a technical invasion may also be uncovered. It appears that international patenting has been an excellent indicator of the industrial rankings of nations, despite the very great differences of patent legislation between those nations, especially prior to the turn of the twentieth century.

Table 1 summarises the world picture of patents granted to the year 1912, which totalled some 3.4 million. Reforms in national patent systems from the 1850s had served to reduce costs and increase security, and this combination had generally driven the vast increase in patenting activity amongst the industrial nations. Thus the reforms in Britain led

TABLE 1  
Patents granted — the World to 1912

| Country                   | Patents   | %    |
|---------------------------|-----------|------|
| USA                       | 1,059,282 | 31.1 |
| France                    | 472,400   | 13.6 |
| Great Britain             | 458,143   | 13.4 |
| British Dependencies (24) | 242,243   | 7.1  |
| Germany                   | 272,069   | 7.9  |
| Belgium                   | 260,159   | 7.6  |
| Austria/Hungary           | 213,950   | 6.3  |
| Italy                     | 115,860   | 3.4  |
| Sub Total                 | 3,094,106 | 90.7 |
| Rest of Europe            | 251,035   | 7.4  |
| Rest of World             | 65,040    | 1.9  |
| (including Japan)         | (23,210)  |      |
| (and Brazil)              | (8116)    |      |
| (and Chile)               | (2540)    |      |
| (and Turkey)              | (2260)    |      |
| Grand Total               | 3,410,185 | 100  |

to the average annual number of accepted patent specifications for 1855–83 reaching 2600, the average for 1884–88 rising to 9500, for 1889–98 to 12,500, and the years 1899–1908 reaching the much higher annual average of 16,000. For the total period 1855–1908 in Britain, some 50% of all patent specifications were in the area of engineering processes, over 10% concerned transport technologies, 8% product technologies, with a less easily-defined miscellany of 30%.

But it will be noted that despite its early-modern origins and its more than modest growth from the 1850s, the British system had not yielded the greatest number of patents, which at over one million easily belonged to the United States, followed by France — it should however be noted that adding the 24 British Dependencies yields a total which outranks that of France. In total, industrial Europe and the USA contributed over 90% of world patenting, the rest of Europe 7%, the rest of the world 2%. Given that Japan dominated the rest of the world with its 23,210 patents, we might conclude on the basis of such very crude statistics that patents granted in very different national systems nevertheless reflected a global industrial reality.

Table 2 provides finer detail for the four major industrial nations and patent systems during 1905–10. Of almost 400,000 patents granted, over half came from the United States, Britain still holding second place in front of Germany and then France. Two points are of more importance as suggesting the value of patent statistics. First, the major industrial nations were significantly ‘open’ ones — over 50% of patents granted in France were to foreigners, this reducing to 44% for Britain, 33% for Germany and a very low 10.5% for the USA. But secondly, with over 25,000 patents lodged overseas, USA patentees showed a level of foreign penetration only just beaten by German patentees. So the two fastest growing industrial nations, ones whose industrial success was associated with the application of new technologies in steel and chemicals (the ‘high-techs’ of that time), were also the greatest lodgers of patented technique into other industrial systems. Clearly, such gross figures

TABLE 2  
Patents granted in major systems, 1905–10

|                          | USA     | UK     | GERMANY | FRANCE* | TOTAL   |
|--------------------------|---------|--------|---------|---------|---------|
| Total Patents Granted    | 204,729 | 93,383 | 71,985  | 29,530  | 399,627 |
| Total to Foreigners      | 21,501  | 41,042 | 23,753  | 15,287  | 101,583 |
| % to Foreigners          | 10.5    | 44.0   | 33.0    | 51.8    | 25.5    |
| of which to USA          | —       | 15,627 | 6424    | 3128    | 25,179  |
| UK**                     | 9503    | —      | 4456    | 2818    | 16,777  |
| Germany                  | 8445    | 13,740 | —       | 5705    | 25,890  |
| France**                 | 1914    | 4713   | 3591    | —       | 10,218  |
| Belgium                  | 335     | 844    | 907     | 604     | 2690    |
| Austro-Hungary***        | 761     | 1532   | 2909    | 544     | 5746    |
| Switzerland              | 579     | 1144   | 1879    | 821     | 4423    |
| Sub Total 6 of 7 Nations | 19,537  | 37,600 | 20,166  | 13,620  | 90,923  |
| % 6 of 7 Nations         | 10      | 40     | 28      | 46      | 22.8    |
| Rest of the World        | 1964    | 3442   | 3587    | 1667    | 10,660  |
| % Rest of World          | 1       | 4      | 5       | 6       | 2.7     |

\* Figures for two years only 1909–10

\*\* Includes dependencies

\*\*\* From 1909 Austria only

reflect an underlying industrial and trading reality, and provide some measure for the expanding world economy.

Of the four systems the two of most contrast were those of the UK and the USA. The American patent system was all but closed to foreign patentees, but in contrast American patentees lodged their patents very aggressively, particularly into Britain and Germany. The British patent system was very open indeed to foreign lodgements, less aggressive in lodging into its industrial competitors, although successful in America — 57% of British foreign lodgements were into the American technological system. Whilst to some extent this pattern was a function of sheer size, it was not a function of patent cost (as shall be seen in the section Patents as Things. . . below). Patenting was far cheaper in the USA than it was in UK. This picture of an American ‘technological invasion’ from a relatively large but closed system seems a fair reflection of early twentieth century economic and commercial history.

The far simpler Table 3 confirms this. Over the years 1904 to 1912, the change in ranking of the four nations in terms of *per capita* patenting does seem to reflect their changed ranks in terms of industrialisation. So the removal of gross population effects suggests that patent statistics prior to 1914 reflected underlying industrial, production and technological realities.

Table 4 refines further by examining patents lodged by foreigners into the British patent system in select years between 1855 and 1870. As such trans-national patenting cost additional money, time and effort and demanded some extra knowledge, the 2062 communications represented an elite component of the patent universe. The left hand column ranks the 10 nations who overwhelmingly patented into Britain via communications

TABLE 3  
Ranking of gross per capita patenting in  
four nations 1904 and 1912

| 1904    | 1912    |
|---------|---------|
| UK      | USA     |
| USA     | Germany |
| Germany | UK      |
| France  | France  |

TABLE 4  
Foreign patents lodged in the UK system in selected years 1855–70

|             | 1855 | %    | 1860 | %    | 1865 | %    | 1870 | %    | Totals | %    |
|-------------|------|------|------|------|------|------|------|------|--------|------|
| USA         | 52   | 31.9 | 247  | 42.8 | 263  | 41.7 | 378  | 54.7 | 940    | 45.6 |
| France      | 85   | 52.2 | 268  | 46.5 | 256  | 40.6 | 181  | 26.2 | 790    | 38.3 |
| Germany     | 5    | 3.1  | 27   | 4.7  | 33   | 5.2  | 30   | 4.3  | 95     | 4.6  |
| Belgium     | 4    | 2.5  | 6    | 1.0  | 15   | 2.4  | 13   | 1.9  | 38     | 1.8  |
| Russia      | 0    | 0    | 1    | 0.2  | 7    | 1.1  | 8    | 1.2  | 16     | 0.8  |
| Canada      | 1    | 0.6  | 3    | 0.5  | 9    | 1.4  | 21   | 3.0  | 34     | 1.7  |
| Italy       | 0    | 0    | 4    | 0.7  | 8    | 1.3  | 3    | 0.4  | 15     | 0.7  |
| Australia   | 0    | 0    | 1    | 0.2  | 5    | 0.8  | 11   | 1.6  | 17     | 0.8  |
| Switzerland | 3    | 1.8  | 5    | 0.9  | 5    | 0.8  | 3    | 0.4  | 16     | 0.3  |
| Spain       | 0    | 0    | 1    | 0.2  | 5    | 0.8  | 1    | 0.1  | 7      | 0.3  |
| Others      | 13   | 8.0  | 14   | 2.4  | 25   | 4.0  | 42   | 6.1  | 94     | 4.6  |
| Total       | 163  |      | 577  |      | 631  |      | 691  |      | 2062   |      |

by their designated agents in these years. USA ranks 1 as providing 46% of foreign patenting in Britain, followed by France with 38%, and then the dramatic fall to 4.61% for Germany, and well below this for all the rest. This table accurately reflects the world of national technological capacities prior to the unification of Germany and Italy. Indeed, excepting the particular institutional and commercial positions of Canada and Australia,<sup>4</sup> the left hand column effectively provides a ranking of per capita industrialisation levels prior to 1870. More particularly it shows a rise through the ranks of the USA, and a fall through the ranks of France.

Table 5 uses the same method applied to a small dependent patent system on the other side of the world. Obviously, colonial relations meant that Britain and the British Dependencies loomed disproportionately as foreign activists in both the colony of New South Wales between 1886 and 1903 and in the united Commonwealth of Australia between 1904 and 1918. Nevertheless, the later period well demonstrates the growing sophistication of international patenting, the aggressive presence of a united Germany, and the emergence of late industrialisers such as Japan and Hungary. Of real significance is the rise of United States patenting in Australia, ousting the British patentees during 1904–18 for a variety of reasons — these including the surge of US corporate patenting that targeted specifically Australian engineering needs, such as ore conveyance, coal cutting machinery, electro-pneumatic channellers, rock and hammer drills and compressors, cyanide gold extraction, and so on.

Table 6 shifts focus to the German patent system after unification, and is designed to show how in the years 1877–94 this relatively new institution reflected the character of industrialisation directly. Amongst the groups of patent created by sub-grouping the 89 categories of the German patent office, chemicals and electricals, motive power, textiles, and mining and metallurgy figured as the largest in that order, a signal measure of the growing industrial economy.

In Table 7 we get closer to the social and information basis of patenting itself. The table shows the industrial character of the 115,000 patent specifications taken out for study at the Manchester patent office library in the years 1857–60.<sup>5</sup> The massive readership of patent specifications for textile fabrics and processes stands out clearly, and is what might be expected from the industrial structure of the city. But notice also the great inventive interest in steam and steam engines and in gas manufacture, as well as the swelling of varying interests in particular years. As has been stressed elsewhere (see section Patents as Guides. . . below), the patent systems acted as both institutions defining technological property rights and networks of engineering expertise and information.

Table 8 returns to Germany and captures the invasion of the British chemical industry by six German organic chemical firms. Of the 948 coal-tar chemistry patents entering the British system from these firms alone in this short period of 1886–1900, 306 originated with Bayer Co. and 231 with Meister, Lucius and Bruning. As may be seen, such aggressive patenting stemmed from the industrial expertise of some 850 engineers and chemists employed in the six German firms. Apart from employing distinguished scientists (Badische had hired Heinrich Caro as its chief chemist in 1868<sup>6</sup>) the employment growth rate for the research divisions of these German firms was extremely fast — Meister, Lucius and Bruning (Hochst) were employing 307 chemists and 74 engineers by 1912.<sup>7</sup> In reply, the six largest British chemical firms patented only 86 chemical patents in their *own* patent system in those years, the leading firms being Read, Holliday and Co. (28 patents), Clayton Aniline Co. (21), and Levinstein (19)! So the corporate foreign lodgements give a direct measure of the impact of the wider German system of the period<sup>8</sup> — by the late 1890s some 4000 trained

TABLE 5  
Foreign activity in the Australian patent system, 1886–1918

| Nations            | Applications in New South Wales |         |         |         |           |         | Applications in Australian Commonwealth |         |         |         |         | RANK      |         |           |  |
|--------------------|---------------------------------|---------|---------|---------|-----------|---------|-----------------------------------------|---------|---------|---------|---------|-----------|---------|-----------|--|
|                    | 1886-88                         | 1889-91 | 1892-94 | 1895-97 | 1898-1900 | 1901-03 | 1904-06                                 | 1907-09 | 1910-12 | 1913-15 | 1916-18 | 1886-1903 | 1904-18 | 1886-1919 |  |
| Australasia        | 1044                            | 1701    | 1509    | 1775    | 1701      | 2028    | 5566                                    | 6558    | 8225    | 8111    | 7513    | 1         | 1       | 1         |  |
| Britain            | 270                             | 413     | 289     | 398     | 418       | 406     | 810                                     | 846     | 1068    | 910     | 825     | 2         | 3       | 2         |  |
| British Deps       | 10                              | 12      | 12      | 33      | 39        | 68      | 166                                     | 197     | 145     | 152     | 79      | 5         | 5       | 5         |  |
| USA                | 96                              | 162     | 160     | 234     | 332       | 408     | 894                                     | 904     | 1121    | 1024    | 921     | 3         | 2       | 3         |  |
| Germany            | 26                              | 21      | 32      | 38      | 39        | 48      | 138                                     | 205     | 292     | 179     | 17      | 4         | 4       | 4         |  |
| France             | 20                              | 17      | 9       | 32      | 45        | 28      | 97                                      | 102     | 124     | 77      | 55      | 6         | 6       | 6         |  |
| Sweden             | 8                               | 6       | 5       | 10      | 7         | 29      | 57                                      | 58      | 56      | 78      | 96      | 7         | 7       | 7         |  |
| Denmark            | 1                               | 1       | 2       | 5       | 23        | 8       | 21                                      | 34      | 24      | 13      | 20      | 8         | 9       | 9         |  |
| Austria            | 2                               | 10      | 3       | 3       | 6         | 8       | 26                                      | 20      | 19      | 12      | 0       | 9         | 14      | 12        |  |
| Belgium            | 6                               | 4       | 1       | 8       | 9         | 3       | 14                                      | 33      | 27      | 24      | 3       | 10        | 11      | 10        |  |
| Russia             | 3                               | 2       | 2       | 2       | 4         | 3       | 4                                       | 17      | 15      | 3       | 0       | 11        | 16      | 16        |  |
| Italy              | 0                               | 0       | 1       | 6       | 2         | 5       | 26                                      | 25      | 15      | 23      | 22      | 12        | 10      | 11        |  |
| Switzerland        | 2                               | 0       | 1       | 1       | 1         | 7       | 20                                      | 21      | 20      | 8       | 13      | 13        | 13      | 13        |  |
| Norway             | 0                               | 0       | 1       | 1       | 1         | 4       | 8                                       | 6       | 14      | 12      | 24      | 14        | 15      | 15        |  |
| Latin & C. America | 2                               | 2       | 0       | 3       | 0         | 0       | 0                                       | 3       | 0       | 10      | 11      | 14        | 17      | 17        |  |
| Netherlands        | 0                               | 0       | 0       | 0       | 5         | 1       | 4                                       | 18      | 6       | 24      | 33      | 16        | 12      | 14        |  |
| Spain and Portugal | 0                               | 2       | 2       | 1       | 0         | 1       | 2                                       | 2       | 4       | 3       | 0       | 16        | 19      | 19        |  |
| Japan              | 0                               | 0       | 1       | 0       | 0         | 0       | 0                                       | 2       | 3       | 6       | 11      | 19        | 18      | 18        |  |
| Elsewhere          | 0                               | 3       | 0       | 0       | 1         | 0       | 13                                      | 21      | 27      | 69      | 55      | 18        | 8       | 8         |  |
| Hungary            | 0                               | 0       | 0       | 0       | 0         | 0       | 0                                       | 0       | 2       | 9       | 0       | 20        | 20      | 20        |  |
| TOTAL              | 1490                            | 2356    | 2030    | 2550    | 2633      | 3055    | 7866                                    | 9072    | 11,204  | 10,747  | 9698    | —         | —       | —         |  |
| % Foreign          | 30                              | 28      | 26      | 30      | 35        | 34      | 29                                      | 28      | 27      | 25      | 23      | —         | —       | —         |  |
| % UK & Deps        | 19                              | 18      | 15      | 17      | 17        | 15      | 12                                      | 11      | 9       | 11      | 9       | —         | —       | —         |  |
| % USA              | 6                               | 7       | 8       | 9       | 13        | 13      | 11                                      | 10      | 10      | 10      | 9       | —         | —       | —         |  |

TABLE 6  
The German patent system, 1877–94; categories of patents granted

| Patent Groups           | No. of sub-groups | Patents Granted 1877–94 |      | Patents Granted 1893–94 |      | % of grants to applications 1893–94 |
|-------------------------|-------------------|-------------------------|------|-------------------------|------|-------------------------------------|
|                         |                   | No.                     | %    | No.                     | %    |                                     |
| Mining and metallurgy   | 11                | 6852                    | 8.6  | 1045                    | 8.2  | 53.1                                |
| Chemical and electrical | 10                | 9128                    | 11.5 | 1809                    | 14.2 | 44.5                                |
| Textiles                | 10                | 6996                    | 8.8  | 1077                    | 8.5  | 45.1                                |
| Agricultural            | 1                 | 2902                    | 3.6  | 463                     | 3.6  | 44.3                                |
| Transport               | 4                 | 4762                    | 5.9  | 751                     | 5.9  | 44.3                                |
| Construction            | 1                 | 1099                    | 1.4  | 131                     | 1.0  | 21.3                                |
| Motive Power            | 8                 | 8792                    | 11.0 | 1025                    | 8.1  | 34.3                                |
| Food and Drink          | 8                 | 5863                    | 7.4  | 835                     | 6.6  | 38.0                                |
| Misc. Manufacturers     | 17                | 20,067                  | 25.2 | 3524                    | 27.7 | 49.1                                |
| Other                   | 19                | 13,158                  | 16.5 | 2050                    | 16.1 | 40.2                                |
| Total                   | 89                | 79,620                  | 100  | 12,710                  | 100  | 43.5                                |

TABLE 7  
References to patent specifications at Manchester

| Subjects                      | 1860  | 1859  | 1858  | 1857  | Total  |
|-------------------------------|-------|-------|-------|-------|--------|
| Total                         | 29241 | 36972 | 27856 | 20877 | 114946 |
| Chemical compositions         | 283   | 1829  | 355   | –     | 2467   |
| Dyeing & colouring            | 625   | 692   | 260   | 257   | 1834   |
| Gas manufacture               | 660   | 8785  | 718   | 264   | 10427  |
| Motive power & propulsion     | 1797  | 221   | 437   | 1243  | 3698   |
| Railways                      | 1154  | 839   | 1295  | 2185  | 5473   |
| Steam and steam engines       | 2834  | 2717  | 3169  | 728   | 9448   |
| Sewing machines               | 2514  | 1100  | 626   | 578   | 4818   |
| Textile fabrics and processes | 9313  | 8414  | 6261  | 1689  | 25677  |
| Telegraphs and signals        | 1865  | 100   | 120   | 503   | 2588   |
| Sub-total                     | 21045 | 24697 | 13241 | 7447  | 66430  |
| % Sub-total                   | 72    | 67    | 47    | 36    | 58     |

chemists were working in the German chemical industry, and by 1900 the German dyestuffs industry represented 90% of the world's total output and was very heavily engaged in the provision of scientific facilities and training.<sup>9</sup>

Table 9 is perhaps the most unusual indicator of the saliency of patent data. Dividing the 142,500 patents filed in the US patent system in 1856–69 into three sub-periods it is possible to very clearly depict the impact of war and its aftermath. Clearly enough, until the outbreak of the Civil War patents averaged at around 5000 plus per annum, were maintained during the war but immediately escalated to an average of 20,000 per annum after the war was over in 1865–69. This is a result uncomplicated by foreign interference — at this time the American system was at its most closed, with less than 4% of patents being granted to foreign residents. More remarkably, the percentage of all agricultural sector patents that



TABLE 8

Scientific enterprise and strategic patenting in the six largest organic chemical firms in Germany, 1886–1900

| Chemical firm                    | Chemist | Engineers/<br>Technologists | All Other<br>Workers | Completed UK Patents,<br>coal-tar 1886–1900 |
|----------------------------------|---------|-----------------------------|----------------------|---------------------------------------------|
| Badische Aniline Works           | 148     | 75                          | 6790                 | 179                                         |
| Meister, Lucius & Bruning        | 120     | 36                          | 3766                 | 231                                         |
| Farbenfabriken                   | 145     | 175                         | 4700                 | 306                                         |
| Berlin Aniline Co.               | 55      | 31                          | 1950                 | 119                                         |
| Cassella & Co.                   |         | 60                          | 1970                 | 75                                          |
| Farbwerk, Muhlheim,<br>Leonhardt | –       | –                           | 450                  | 38                                          |

TABLE 9

The US patent system and Civil War, 1861–65

|         | Patents Filed | Patents Issued |
|---------|---------------|----------------|
| 1856–60 | 28973         | 18299          |
| 1861–65 | 33291         | 22667          |
| 1866–69 | 80236         | 49829          |

arose in the Southern slave states before the civil war was around 2.5% of the total. By 1869 that figure had risen to 23%. Where before the civil war the industrial north had provided the agricultural technologies for the slave-owning South, after the civil war and Lincoln's Emancipation of 1 January 1863, the so-called 'new system of labour' imposed on the South meant that agricultural invention moved dramatically towards the Southern agricultural states as improved machinery was sought to replace slave labour.

A final vignette is provided in Table 10, which shows a breakdown of the foreign origins of Japanese patents in the years 1896–1904. What is particularly noticeable is that during

TABLE 10

Foreign applications and registrations in Japan, November 1896–December 1904

|                    | PATENTS               |                                      | DESIGNS               |                                      | TRADE MARKS           |                                      |
|--------------------|-----------------------|--------------------------------------|-----------------------|--------------------------------------|-----------------------|--------------------------------------|
|                    | No. Appli-<br>cations | Registrations<br>% Appli-<br>cations | No. Appli-<br>cations | Registrations<br>% Appli-<br>cations | No. Appli-<br>cations | Registrations<br>% Appli-<br>cations |
| Total              | 18616                 | 29                                   | 5327                  | 32                                   | 24058                 | 64                                   |
| (all Japan)        |                       |                                      |                       |                                      |                       |                                      |
| All Foreign        | 1938                  | 65                                   | 25                    | 50                                   | 4018                  | 76                                   |
| UK                 | 442                   | 64                                   | 12                    | 75                                   | 1731                  | 74                                   |
| USA                | 973                   | 71                                   | 1                     | 100                                  | 532                   | 77                                   |
| Germany            | 289                   | 47                                   | 7                     | 14                                   | 1332                  | 78                                   |
| Total %<br>Foreign | 10                    | –                                    | 0.04                  | –                                    | 17                    | –                                    |
| Total % USA        | 5                     | –                                    | 0.002                 | –                                    | 2                     | –                                    |

that nation's early industrialisation period, foreign patenting remained very low, at around 10% of the relatively small national total. On the other hand, foreign activity in registering trade-marks in Japan was quite high. This pattern of patents reflects directly the fact that imports of foreign capital were restricted by law, although foreign trade was welcomed and thriving. This made patenting in Japan relatively unattractive to foreigners, but encouraged German and American manufacturers in particular to register their trademarks in a fairly fruitless attempt to prohibit complete imitation of their products by Japanese manufacturers or merchants.<sup>10</sup> On the other hand, Japanese industrialisation in the Meiji years appears to have been based on a process of technological borrowing that was more dependent on trade than on investment.<sup>11</sup>

### PATENTS AND SITES OF INVENTIVE ACTIVITY

It is quite common amongst historians of engineering and technology to hear conversations claiming that particular sites or places, or even types of place, are more inventive or innovative than others. Thus we might debate whether the Golden Ages of great cities were also their times of maximum technical innovative activity, or, contrariwise, whether the energy, attention and consumer demand devoted to aesthetics and the arts displaced technical achievements.<sup>12</sup> Clearly, this becomes an even broader and more heated political discussion amongst policy makers in nations such as the US, Japan, Germany and Britain at the present time, nations who now spend many billions in total on attempting to 'incubate' the best of technological investigations and outcomes, often in very costly science cities and technology parks.<sup>13</sup> In this paper a focus on Britain allows an identification of innovative sites in some detail. In particular the question of the supposed dominance of London may be addressed. Those who might wish to minimise the importance of patent statistics would argue that the seeming predominance of London in the figures reflects the importance of the siting of the patent offices and agents in London, and that inventors situated any great distance from London were less likely to patent (because of monetary or opportunity costs), a measure of the minimal value that they themselves placed on patenting as either an egoistical, a commercial or a protective strategy. The data challenges this claim directly and shows strong alignments between sites of patenting and sites of industrial growth.

The rather fulsome Table 11 addresses this question directly by apportioning its 5109 patents for the 1840s between six industrial counties and the eight 'Home' counties proximate to London. It will be seen that the combined Home Counties contributed only 298 or 5.8% patents to the 1840s total. Much further from London, and so involving higher money and opportunity cost, the combined industrial counties contributed 1389 or 27% of the total, falling behind London itself at 1977 or 38% of the total for this decade. So at the end of the patent *ancient régime*, non-industrial areas proximate to London patented far less than more distant industrial counties. The exceptional character of London itself is uncovered below. A second feature of the table is the emergence of engineer-patents.<sup>14</sup> Of the total patentees, 26% were engineers, the proportions for London being 26%, the industrial counties 28%, and the Home Counties 25%. The variations are insignificant and it may be concluded that the pre-reform system was characterised by industrial-region patenting, primarily by artisan groupings applying industry-based skills. More general engineering applications were found across the board — even the proportion of foreign patentees that were engineers came to 26%.

TABLE 11  
Geographical location of British patenting, 1840-49

|                           | 1840 | 1841 | 1842 | 1843 | 1844 | 1845 | 1846 | 1847 | 1848 | 1849 | Total | All Engineers |
|---------------------------|------|------|------|------|------|------|------|------|------|------|-------|---------------|
| All Patentees             | 491  | 508  | 409  | 472  | 524  | 587  | 541  | 539  | 447  | 591  | 5109  | 1318          |
| All Patents               | 439  | 440  | 371  | 420  | 450  | 572  | 493  | 493  | 288  | 514  | 4480  |               |
| London                    | 196  | 204  | 172  | 184  | 194  | 242  | 214  | 222  | 158  | 191  | 1977  | 523           |
| Middlesex                 | 130  | 138  | 121  | 134  | 124  | 143  | 137  | 144  | 93   | 128  | 1292  | 353           |
| Edinburgh/Glasgow         | 9    | 15   | 3    | 5    | 10   | 10   | 9    | 6    | 5    | 11   | 83    | 25            |
| USA                       | 1    | 0    | 0    | 0    | 2    | 8    | 4    | 10   | 7    | 7    | 39    | 8             |
| France                    | 10   | 4    | 0    | 4    | 8    | 16   | 43   | 25   | 10   | 29   | 149   | 42            |
| Belgium                   | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 0    | 4    | 0    | 5     | 0             |
| No. of Foreign Patents    | 11   | 4    | 0    | 4    | 11   | 24   | 47   | 35   | 21   | 36   | 193   | 50            |
| Lancashire                | 68   | 56   | 55   | 50   | 64   | 68   | 62   | 72   | 81   | 93   | 669   | 225           |
| Nottinghamshire           | 4    | 7    | 4    | 6    | 17   | 6    | 7    | 8    | 4    | 7    | 70    | 17            |
| Cheshire                  | 0    | 0    | 3    | 0    | 1    | 1    | 0    | 0    | 3    | 0    | 8     | 1             |
| Staffordshire             | 17   | 12   | 7    | 10   | 14   | 7    | 12   | 17   | 7    | 10   | 113   | 17            |
| Yorkshire                 | 27   | 15   | 20   | 17   | 20   | 25   | 17   | 22   | 15   | 36   | 214   | 70            |
| Warwickshire              | 37   | 41   | 29   | 26   | 28   | 27   | 37   | 24   | 32   | 34   | 315   | 62            |
| Total Industrial Counties | 153  | 131  | 118  | 109  | 116  | 107  | 135  | 119  | 142  | 146  | 1389  | 392           |
| Hertfordshire             | 1    | 1    | 2    | 0    | 0    | 4    | 0    | 1    | 0    | 2    | 11    | 1             |
| Essex                     | 0    | 4    | 3    | 4    | 3    | 3    | 3    | 5    | 5    | 6    | 36    | 7             |
| Surrey                    | 12   | 16   | 13   | 30   | 23   | 30   | 15   | 20   | 15   | 23   | 197   | 54            |
| Berkshire                 | 1    | 1    | 1    | 0    | 0    | 1    | 2    | 1    | 2    | 1    | 10    | 4             |
| Buckinghamshire           | 5    | 0    | 1    | 3    | 4    | 1    | 3    | 1    | 2    | 1    | 21    | 1             |
| Oxfordshire               | 0    | 1    | 0    | 1    | 0    | 1    | 0    | 0    | 1    | 0    | 4     | 6             |
| Hampshire                 | 0    | 1    | 0    | 0    | 0    | 0    | 0    | 1    | 0    | 0    | 2     | 1             |
| Bedfordshire              | 5    | 2    | 4    | 0    | 0    | 0    | 2    | 2    | 0    | 2    | 17    | 1             |
| Total Home Counties       | 24   | 26   | 24   | 38   | 30   | 40   | 25   | 31   | 25   | 35   | 298   | 75            |

Table 12 simply lists the urban patenting of the years 1830–49. Notably, such urban patenting covered 74% of the total of 7,410 patents. It is important to note that in these years there was only a small proportion of foreign patentees (5%) recorded as patenting from overseas — most travelled into London for such lodgement and were thus recorded as metropolitan-based rather than foreign. London, of course, stands out with a huge 61% of the urban total, followed by Birmingham (with 8%), Manchester (with 6%), Liverpool (with 2.6%) and Nottingham (with 2.5%). It is worth noting that of all world cities, Parisian patenting in Britain equalled Glaswegian and outnumbered that of Leeds, Leicester or Edinburgh.

Table 13 moves beyond the patent reform period to the years 1855–70. Categorising 14,559 direct patentees, it follows patent office practice and lists foreign patentees as only those communicating from overseas, assigning foreign patentees temporarily resident in London as metropolitan. In these years the great bulk of British inventive activity — as measured in patent data — took place in London and the industrial counties. Scotland contributed three times as many patents as the combined Home Counties but less than 18% of the six English industrial counties. Even defining foreign patenting as embracing only foreign communications, this total roughly matched the combined patenting total of the Home Counties, Scotland, Ireland and Wales. The remaining 14% of patentees were distributed amongst the other English counties.

More ambitiously, Table 14 ranks all British urban patenting from 1700 to 1881. The top ten or so cities show a great consistency. It may be shown quite quickly that this is primarily a function of size rankings by re-ranking the urban areas in ‘patents per capita’ terms for 1852–70, as is done in Table 15. The results are significant and immediate. When size of population is accounted for, as in column 3, the highest ranked cities become Rochdale, Nottingham, Birmingham, London, Manchester and Halifax, with Liverpool, for example, slipping down into seventeenth place. Patents are now measuring true sites of inventive activity in the sense that they capture intensity or density of effort.

Charts 1, 2, 3 and 4 perform this operation for the industrial counties of Lancashire and Yorkshire. Chart 1 shows that Lancashire was in gross terms dominated by Manchester patenting, followed by Liverpool, Rochdale and Bolton. In contrast, Chart 2 shows that in Yorkshire gross inventive activity was shared by the cities of Leeds, Sheffield, Bradford and Halifax. This seems quite an accurate measure of the different industrial and urban structures of the two counties, but is distorted by sheer size. Chart 3 shows Lancashire urban patenting in per capita terms, and now Manchester and Liverpool shrink in importance, and Accrington, Rochdale and Bolton appear as significant centres of inventive activity. Similarly, Chart 4 demonstrates that in real per capita terms Yorkshire patenting was most intensely sited in Keighley, Halifax, Bradford, Huddersfield and Sheffield.

It may be concluded that inventive activity as measured in patenting was predominantly sited in large urban cities in industrial regions and in London. The ranking of such urban sites is fundamentally altered when patents are enumerated per capita. This suggests that the economies generated by site specialisation in the largest cities, the advantages accorded to such cities through the existence there of more sophisticated and extensive network of information, facilities for information communication and discussion, were not quite large enough to offset simpler population effects. If such large city advantages had been stronger, the switch to per capita calculation would not have had quite such a dramatic effect. Cities that seem to perform well under both calculations, however, include London (falling from 1 to only 4 despite its vast size), Nottingham (shifting decidedly upwards to 2),

TABLE 12  
British patenting in major cities, 1830–49

| Location                | No. of Patentees |
|-------------------------|------------------|
| Accrington              | 7                |
| Beverley                | 5                |
| Birmingham              | 439              |
| Bolton                  | 45               |
| Bradford                | 46               |
| Brighton                | 14               |
| Bristol                 | 41               |
| Bury                    | 30               |
| Bath                    | 15               |
| Deptford                | 4                |
| Durham                  | 57               |
| Edinburgh               | 55               |
| Glasgow                 | 108              |
| Halifax                 | 14               |
| Huddersfield            | 22               |
| Hull                    | 24               |
| Keighly                 | 0                |
| Leeds                   | 87               |
| Leicester               | 94               |
| Liverpool               | 150              |
| London/Middlesex        | 3532             |
| Manchester              | 328              |
| Newcastle               | 55               |
| Nottingham              | 139              |
| Oldham                  | 28               |
| Preston                 | 10               |
| Rochdale                | 29               |
| Salford                 | 58               |
| Sheffield               | 52               |
| Wakefield               | 4                |
| York                    | 10               |
| Belfast                 | 6                |
| Dublin                  | 40               |
| New York                | 18               |
| Paris                   | 107              |
| Other Foreign           | 107              |
| Total Foreign Patentees | 278              |
| Total Above             | 5780             |
| Remainders              | 1630             |
| Total Patentees         | 7410             |

TABLE 13  
Direct patentees in Britain 1855, 1860, 1865 and 1870

|                     | 1855 | 1860 | 1865 | 1870 | Total | % System |
|---------------------|------|------|------|------|-------|----------|
| London              | 1152 | 1348 | 1343 | 1503 | 5346  | 37       |
| Industrial Counties | 1042 | 1073 | 1062 | 1263 | 4440  | 30       |
| Home Counties       | 47   | 38   | 78   | 95   | 258   | 2        |
| Other England       | 470  | 511  | 542  | 538  | 2061  | 14       |
| Scotland            | 163  | 165  | 164  | 294  | 786   | 5        |
| Ireland             | 73   | 65   | 51   | 40   | 229   | 2        |
| Wales               | 17   | 42   | 43   | 52   | 154   | 1        |
| Foreign             | 492  | 343  | 254  | 196  | 1285  | 9        |
| Total               | 3456 | 3585 | 3537 | 3981 | 14559 | 100      |

TABLE 14  
The urban ranking of British patents, 1700–1881

|              | 1700–1858 | 1852–53 | 1855–70 | 1881 |
|--------------|-----------|---------|---------|------|
| London       | 1         | 1       | 1       | 1    |
| Manchester   | 2         | 2       | 2       | 2    |
| Birmingham   | 3         | 3       | 3       | 3    |
| Glasgow      | 4         | 4       | 4       | 4    |
| Liverpool    | 5         | 5       | 5       | 5    |
| Leeds        | 6         | 7       | 6       | 6    |
| Sheffield    | 7         | 8       | 7       | 7    |
| Bristol      | 8         | 14      | 16      | 13   |
| Bradford     | 9         | 6       | 8       | 8    |
| Nottingham   | 10        | 11      | 7       | 9    |
| Edinburgh    | 11        | 10      | 11      | 10   |
| Newcastle    | 12        | 13      | 9       | 14   |
| Dublin       | 13        | 16      | 12      | 18   |
| Rochdale     | 14        | 8       | 13      | 26   |
| Leicester    | 15        | 17      | 15      | 21   |
| Bolton       | 16        | 15      | 10      | 11   |
| Halifax      | 17        | 12      | 15      | 16   |
| Oldham       | 18        | –       | 14      | 17   |
| Huddersfield | 19        | –       | 17      | 15   |
| Derby        | 20        | –       | 18      | –    |

Birmingham (remaining at 3 under both circumstances), Leicester (shifting upwards from rank 15 to rank 13), Newcastle (remaining around the 12–15 mark), Sheffield and Leeds (shifting only from positions 6 and 7 to positions 11 and 12), and Rochdale itself which moved from a low of 13 in gross terms to first ranked in per capita terms. Large cities that boasted buoyant scientific and technical cultures but where such assets did not appear to be reflected in per capita invention, include Liverpool and Bristol, cities of mercantile commerce with strong intellectual elites. It may be proposed that in such urban places the interaction between elite associations and facilities and the sites of invention was far less than in cities such as Rochdale, Nottingham, Birmingham, Sheffield, and even London.<sup>15</sup>

TABLE 15  
Number of patents involving each city/town and ranking, 1852–70

|               | Population | No. Patents All Years | Patents per 1000 population | Rank |
|---------------|------------|-----------------------|-----------------------------|------|
| Rochdale      | 29,195     | 68                    | 2.33                        | 1    |
| Nottingham    | 57,407     | 130                   | 2.26                        | 2    |
| Birmingham    | 232,841    | 492                   | 2.11                        | 3    |
| London        | 2,362,236  | 4900                  | 2.07                        | 4    |
| Manchester    | 303,382    | 592                   | 1.95                        | 5    |
| Halifax       | 33,582     | 57                    | 1.70                        | 6    |
| Huddersfield  | 30,880     | 43                    | 1.39                        | 7    |
| Bradford      | 103,778    | 129                   | 1.24                        | 8    |
| Bolton        | 61,171     | 75                    | 1.23                        | 9    |
| Oldham        | 52,820     | 63                    | 1.19                        | 10   |
| Leeds         | 172,270    | 174                   | 1.01                        | 11   |
| Sheffield     | 135,310    | 130                   | 0.96                        | 12   |
| Leicester     | 60,584     | 57                    | 0.94                        | 13   |
| Glasgow       | 329,097    | 299                   | 0.91                        | 14   |
| Newcastle     | 87,784     | 76                    | 0.87                        | 15   |
| Derby         | 40,609     | 26                    | 0.64                        | 16   |
| Liverpool     | 375,955    | 208                   | 0.55                        | 17   |
| Dublin        | 146,778    | 73                    | 0.50                        | 18   |
| Bristol       | 137,328    | 52                    | 0.38                        | 19   |
| Edinburgh     | 222,015    | 74                    | 0.33                        | 20   |
| Paris, France | 1,053,000  | 593                   | 0.56                        | —    |

A principal site of activity was the areas of Holborn, Fleet Street and the Strand, in very close proximity to the London patent office. Table 16 shows that this area contributed some 40% of all patenting in Britain around 1853, but that this proportion had fallen to 22% by 1881. Whereas at the beginning of the reformed patent system London had represented 70% of the patenting of the top 10 cities and 155% of the patenting of the six industrial counties, by 1881 London patenting had fallen to 57% and 89% respectively. The explanation is that an illusion is being uncovered.

At the beginning of the period the vast majority of foreign patentees lodged their patents directly, by residing temporarily in London and at times using a London-based patent agent. The vast majority of such patentees were thereby recorded as sited in London, and together with sheer size effects (above) this greatly swelled the London figures. By the 1860s or 1870s, the new legislations had encouraged foreign patenting by foreign communications, whereby the foreign patentee remained overseas and almost always used a London-based agent in order to secure a British patent. In the great majority of cases these patentees were recorded as genuinely foreign and thus the great fall in the seeming predominance of the British metropolis. Table 17 confirms this interpretation by giving details of patenting in Holborn in 1855–70. Of over 1000 patentees recorded for Holborn during the selected years, some 80% were patent agents or engineer-agents, almost all of whom depended for their livelihood on re-writing, advising and managing the patent communications of overseas innovators. In summary, the switch of foreign patentees from direct British patenting to foreign patenting by communication removed the illusion of London's great prominence in

CHART 1

% patentees in Lancashire 1855, 1860, 1865 and 1870

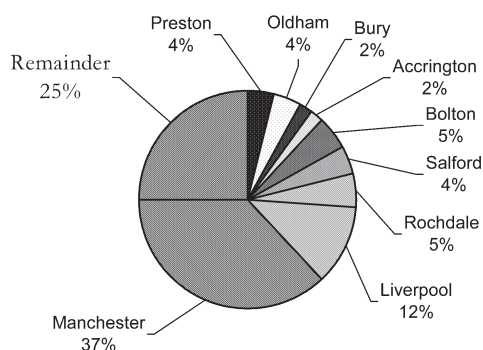


CHART 2

% patentees in Yorkshire 1855, 1860, 1865 and 1870

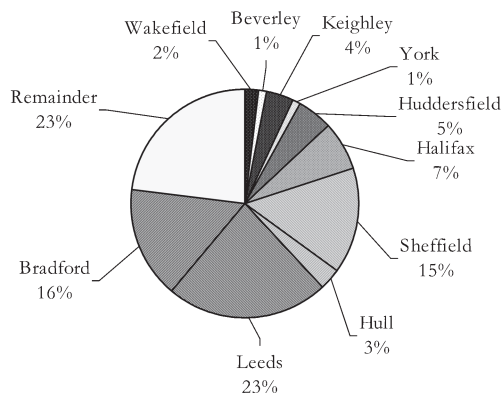


CHART 3

Lancashire – patentees per 1000 population

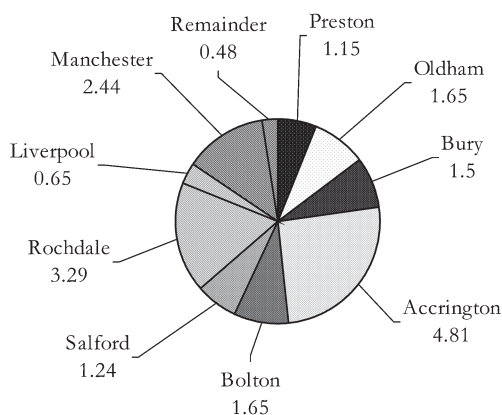
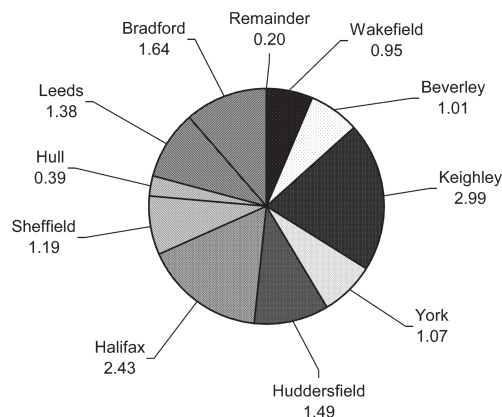


CHART 4

Yorkshire patentees per 1000 population



British patenting. The ‘per capita effect’ serves to further reduce the weight of London. Together with the data that shows the low level of patenting from the eight Home Counties closest to the London patent office, this serves to destroy any thesis that British patenting was sensitive to immediate cost or difficulty differences. Indeed, the sites of British patenting were more commonly those of medium-sized industrial cities. This finding is further boosted with material showing the character of the inventor’s information system (see section ‘Patents as Guides. . .’ below) and the relatively high cost of British patenting even after 1851–52 (see section ‘Patents as Things. . .’ below).

## PATENTS AS GUIDES TO INVENTORS AND THEIR WORLDS

Much has been asked of the character of the nineteenth-century inventor or innovator. How many were ‘self-made’? How was invention financed? How was knowledge diffused and



TABLE 16  
London patent applications in the British system, 1852–53 and 1881

|                                        | 1852–53 | 1881  |
|----------------------------------------|---------|-------|
| Total Patents                          | 4297    | 5751  |
| London Patents % Grand total           | 42.66   | 21.91 |
| London Patents % British Total         | 46.48   | 34.68 |
| London Patents % Top 10 UK Cities      | 69.67   | 57.32 |
| London Patents % Top 20 UK Cities      | 64.16   | 51.43 |
| London Patents % 6 Industrial Counties | 154.68  | 88.79 |
| Foreign Patents % Grand Total          | 8.22    | 37.19 |

TABLE 17  
Occupations of Patentees in Holborn, London, 1855–70

|                                             | 1855 | 1860 | 1865 | 1870 | Total |
|---------------------------------------------|------|------|------|------|-------|
| Patent Agent                                | 34   | 100  | 117  | 88   | 339   |
| Engineer/Patent Agent                       | 46   | 189  | 250  | 249  | 734   |
| Engineer                                    | 3    | 12   | 11   | 19   | 45    |
| Manufacturer                                | 12   | 8    | 5    | 6    | 31    |
| Artisan Tradesman                           | 4    | 6    | 3    | 2    | 15    |
| Merchant                                    | 1    | 0    | 2    | 2    | 5     |
| Retail Tradesman                            | 0    | 3    | 2    | 0    | 5     |
| Professional                                | 2    | 0    | 2    | 4    | 8     |
| Commercial/Clerk                            | 0    | 0    | 0    | 1    | 1     |
| Other                                       | 2    | 0    | 3    | 0    | 5     |
| Gentlemen/Esquire                           | 8    | 7    | 6    | 2    | 23    |
| Unknown                                     | 25   | 21   | 32   | 64   | 142   |
| Total                                       | 137  | 346  | 433  | 437  | 1353  |
| Holborn as % of London                      | 12   | 26   | 32   | 28   | 25    |
| % Holborn Patentees acting as Patent Agents | 80   | 84   | 85   | 77   | 79    |

brought to bear on particular technical problems? What was the link between technical innovators and entrepreneurs, and so on? Within this sort of query is the more conceptual one relating to the links between specific industry skills (as in say trade apprenticeships) and more general, theoretical knowledge and expertise, as perhaps gained in engineering training and formalised education.<sup>16</sup> Rosenberg has helped re-specify this query with his well-known arguments relating to the increasing importance through time of the capital goods industries (the site of engineering and engineers) as the most aggressive centres of technical innovation, eventually replacing older trade skills and workshops.<sup>17</sup> The evidence here gives a mixed result and complicates the picture.

Table 18 provides both an occupational and a regional breakdown of the 5000 or so patentees of the 1840s. Across the board, engineers represent around one quarter of the total, tradesmen and artisans about 36%, manufacturers around 10% (although much higher in the industrial counties), and the ubiquitous ‘gentleman’ retains a 22% rate<sup>18</sup> (although much lower in the industrial counties). It is clear that inventive activity was based

TABLE 18  
Occupation/location of British patenting, 1840–49

| Occupation/Location | Total Industrial Counties | Total Home Counties | Patentees in All Areas |
|---------------------|---------------------------|---------------------|------------------------|
| Civil Engineer      | 40                        | 17                  | 335                    |
| Other Engineers     | 203                       | 35                  | 666                    |
| Mechanic            | 25                        | 2                   | 96                     |
| Machinist           | 46                        | 4                   | 84                     |
| Chemist/Chemistry   | 67                        | 16                  | 211                    |
| Manufacturer        |                           |                     |                        |
| Wright              | 12                        | 2                   | 23                     |
| Foreman             | 2                         | 0                   | 7                      |
| Total of Engineers  | 392                       | 75                  | 1318                   |
| Gentlemen/Esquire   | 143                       | 87                  | 1127                   |
| Merchant            | 52                        | 8                   | 276                    |
| Manufacturer        | 249                       | 26                  | 537                    |
| Farmer/Yeoman       | 8                         | 1                   | 34                     |
| Total Occupations   | 844                       | 197                 | 3292                   |
| Above               |                           |                     |                        |
| All Other           | 545                       | 102                 | 1817                   |
| Total Patentees     | 1389                      | 298                 | 5109                   |

on industrial skills and interests, especially in the industrial counties. Investigation within industrial counties confirms a link between the character of industrial patenting and the nature of the occupations of patentees. So, of all patents registered from Nottinghamshire in the 1830–50 period, some 68% were for textile machinery and equipment improvement, of which 55% were in the lace industry. Of all Nottinghamshire patentees, 38% were engineers and machine makers, 42% were textile manufacturers (of whom 66% were lace manufacturers), and 10% were artisans, of whom 70% were employed in textiles. Historical examination of the county shows that the majority of the Nottinghamshire textile manufacturers of the 1830s and 1840s had been employed in the textile trades in earlier years, especially as lace makers, framework knitters and lace-machine mechanics. From this it would seem that in an industrial county such as Nottingham, tacit industrial skill was basic to inventive efforts, the locus probably shifting from artisan/tradesman to small manufacturers mostly through a process of upward mobility of innovative individuals.

Chart 5 showing the occupations of patentees in Britain between 1855 and 1870 is based on an enumerated 10,533 known cases. What should be immediately obvious is the rise of the engineers to dominant first place at 38% of the total. Although artisans and manufacturers have been firmly differentiated in the chart, it might be recognised that the movement between such categories was very fluid, as in the case of Nottinghamshire above. Table 19 confirms the rise of the engineers through the period, to the high point of 42% in 1870.

An expression of the occupational distribution in terms of characteristic regions is provided in Table 20, which divides the population of 7550 cases between the study's six industrial counties, the eight Home Counties, and London. Very noticeable is the fairly equal distribution of the engineers (at around 39%), the definite bias towards the artisan tradesmen in the industrial counties (12.2%), and the massive bias towards manufacturers in the industrial counties (29.9%). As would now be expected, professional patent agents were

CHART 5  
Occupations of patentees in Britain 1855, 1860, 1865  
and 1870

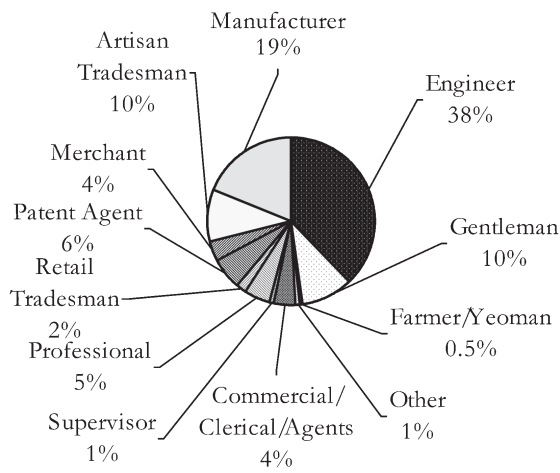


TABLE 19  
% Engineers as patentees 1855–70

|                             | 1855  | 1860  | 1865  | 1870  | Total |
|-----------------------------|-------|-------|-------|-------|-------|
| Number of occupations given | 2577  | 2820  | 2525  | 2738  | 10660 |
| Engineers                   | 843   | 1019  | 1015  | 1159  | 4036  |
| % Engineers                 | 32.71 | 36.13 | 40.20 | 42.33 | 37.86 |

TABLE 20  
% Occupations of patentees by characteristic regions 1855–70

|                        | Industrial Counties | Home Counties | London |
|------------------------|---------------------|---------------|--------|
| Engineer               | 37.8                | 39.4          | 41.7   |
| Manufacturer           | 29.9                | 8.5           | 8.9    |
| Artisan Tradesman      | 12.2                | 10.6          | 6.5    |
| Merchant               | 3.8                 | 0.5           | 3.3    |
| Patent Agent           | 0.5                 | 0             | 17.2   |
| Retail Tradesman       | 2.2                 | 5.9           | 1.5    |
| Professional           | 2.0                 | 8.0           | 5.0    |
| Supervisor             | 2.3                 | 1.1           | 0.3    |
| Commercial/Clerk/Agent | 5.2                 | 1.6           | 1.5    |
| Farmer/Yeoman          | 0.2                 | 4.3           | 0      |
| Gentleman/Esquire      | 3.1                 | 18.1          | 13.1   |
| Other                  | 0.9                 | 2.1           | 1.1    |

TABLE 21  
Occupations of Birmingham patentees 1840–49

|                                       | No. Patentees | %       |
|---------------------------------------|---------------|---------|
| Total No. of Engineers                | 62            | 19.8    |
| Gentleman/Esquire                     | 25            | 8.0     |
| Merchant                              | 16            | 5.1     |
| Manufacturer                          | 81            | 25.8    |
| Farmer/Yeoman                         | 4             | 1.2     |
| (Sub Total)                           | (188)         | (c. 60) |
| Others (mainly<br>Tradesmen/Artisans) | 126           | 40.1    |
| Total                                 | 314           | 100     |

all but confined to London. Gentlemen persisted in the Home Counties but had almost disappeared in the industrial regions.

Such generalisations are refined through a return to Birmingham. Table 21 offers an occupational breakdown of the 314 Birmingham patentees of the 1840s, and shows the great predominance of tradesmen/artisans at around 40%, with manufacturers at 25% and engineers at around 20%. The post-reform system is shown for Birmingham in Chart 6 for 1855–70, covering 593 cases. In a city of very many small metal-working firms, 57% of patenting stemmed from manufacturers, 22% from engineers. Almost certainly, the fall of the artisan/tradesmen to the third rank was a partial function of social mobility, as erstwhile tradesmen became small businessmen through patenting and its ancillary partnerships. That Birmingham did not merely represent urban industrialism is shown by comparison with Chart 7, showing occupations of Manchester patentees in the same period. Here engineers (at 47%) take the place of small manufacturers (20%), this being almost certainly a reflection of the greater role of large firms and capital goods industries in that city. The contrast suggests that technical innovations originated in very particular sites within Britain, with smaller new-product improvements still coming overwhelmingly from industry skills of tradesmen and small manufacturers.

CHART 6

Occupations of patentees in Birmingham 1855–70

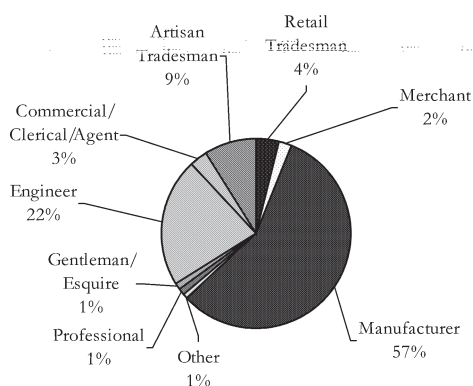


CHART 7

Occupations of patentees in Manchester 1855–70

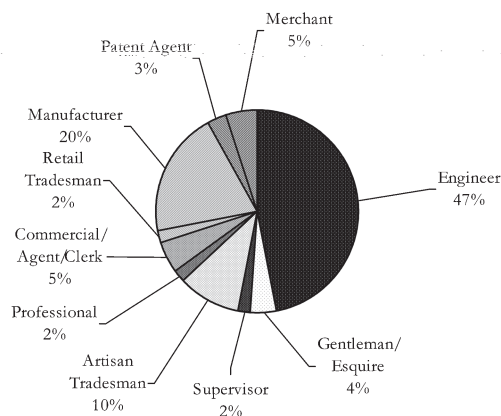


TABLE 22  
Partnerships — number of patents 1855–70

| No. of Partners                      | 1855  | 1860  | 1865  | 1870  | Total Partnerships |
|--------------------------------------|-------|-------|-------|-------|--------------------|
| 2                                    | 380   | 408   | 374   | 496   | 1658               |
| 3                                    | 47    | 7     | 70    | 75    | 199                |
| 4                                    | 7     | 1     | 6     | 7     | 21                 |
| 5                                    | 0     | 0     | 0     | 1     | 1                  |
| 6                                    | 0     | 0     | 2     | 1     | 3                  |
| Total Patents involving Partnerships | 434   | 416   | 452   | 580   | 1882               |
| Total all Patents                    | 2958  | 3196  | 2986  | 3305  | 12445              |
| % of all Patents Partnerships        | 14.67 | 13.02 | 15.14 | 17.55 | 15.12              |
| Partnerships involving Engineers     | 169   | 161   | 171   | 227   | 728                |
| % All Patents                        | 5.71  | 5.04  | 5.73  | 6.87  | 5.85               |
| % Partnerships                       | 38.94 | 38.70 | 37.83 | 39.14 | 38.68              |

Further insight is provided by patent partnerships in this period. Table 22 outlines details for the 1882 patent partnerships in Britain during the select years 1855–70. It is clear that the tendency towards partnership was a growing one and especially involved the engineers — 15% of all patenting was through partnerships, of which 39% involved engineers. This to an extent involved a continuation of pre-reform patent trends, but with some increase in the presence of engineers, over-lookers, foremen and managers. There was an increase in patenting by owners of established firms in partnership with their tradesmen and engineers, but also the emergence of truly corporate patenting (with the patent in the name of the firm), especially in London, Bradford, Sheffield, Manchester, Preston, Birmingham and Edinburgh. Much of this was strategically incremental — the patenting of further modifications to earlier patents held by the firm, ones that had been instrumental in the original formation of the firm. Foreign partnerships were not yet of great significance save those of the ‘gentlemen’, the exception to this being the Parisian chemists and engineers.

Just prior to the reform of the patent system it had become very clear that efficient innovations depended on an improved information system, this especially relating to the vexed question of whether a patent was ‘previously known’ by publication, this prompting the exclamation of Lord Chief Justice Gibbs in the case of Wood versus Zimmer that ‘some inventions almost baffle discovery’. Scientific evidence was clearly liable to defect because it often stemmed from ‘depositions of opinions from persons who have received *ex parte* instructions’.<sup>19</sup> As one agent concluded, a patent specification must be proof against *both* ‘persons of ability, science and learning’ as well as ‘men intimately acquainted with the manufacture, but totally unskilled in science, or learning’.<sup>20</sup> Immediately after the reform it was acknowledged that the new legislation alleviated the financial burden on the artisan of securing provisional protection, but in the absence of informed decisions there was still little facility for enforcing patent rights. Information and rights went hand in hand. Arguing that the legal mind ‘is altogether at fault in following an inference in mechanics or chemistry, or detecting and exposing a sophistry therein’, the agent William Newton lobbied for a special patent court to replace the confused complexity surrounding the Courts of Queen’s Bench, the Exchequer, and Common Pleas, to be paid for through the sales of patent literature ‘in the chief manufacturing towns’.<sup>21</sup> Given the social profile of British patenting after the

reforms of the 1850s, consideration can be given to the character of the information system that could be drawn upon by tradesmen, small manufacturers and engineers. Table 23 shows that patent agency remained centred very much on London, and served primarily as an information system for metropolitan patentees and for foreign lodgements by communications — thus French and other foreign patentees were using London-based agents rather than agencies in their own countries.<sup>22</sup> Urban industrial patentees were thus left to their own devices in more ways than one! Upon what institutions could they rely in order to reduce their opportunity or transaction costs?

The first tier of recourse was that provided in the patent system itself — the cost of printing patent specifications and other literature ran at around a £30,000 annual average, with the total for 1853–58 amounting to £191,442. This represented an enormous public investment in specific information. In the years 1855–79 some 400,000 readers utilised the free public library of the London Patent Office, and in those same years over 4.5 million visitors entered the Patent Office Museum. Outside of London, by 1870 identical information was freely available in 99 provincial libraries and associations, each of which had freely received the complete series of Patent Commissioners' volumes of patent specifications and related material (of which there were 3820 volumes) at a cost of £3900 per batch. Table 24 shows that when to this is added material freely dispersed to colonies, learned

TABLE 23  
Patents involving a patent agent 1855–70

|                     | 1855  | 1860 | 1865  | 1870  | Total | %     |
|---------------------|-------|------|-------|-------|-------|-------|
| London              | 263   | 414  | 467   | 447   | 1591  | 84.63 |
| Industrial Counties | 4     | 10   | 9     | 31    | 54    | 2.87  |
| Other Provincial    | 2     | 5    | 1     | 4     | 12    | 0.64  |
| Paris               | 3     | 0    | 5     | 18    | 26    | 1.38  |
| Other Foreign       | 9     | 0    | 0     | 0     | 9     | 0.48  |
| London and Glasgow  | 82    | 7    | 0     | 19    | 108   | 5.74  |
| London and Foreign  | 0     | 2    | 29    | 49    | 80    | 4.26  |
| Total               | 363   | 438  | 511   | 568   | 1880  |       |
| As % of all Patents | 12.27 | 13.7 | 17.11 | 17.19 | 15.12 |       |

TABLE 24  
The British patent system and the diffusion of technological information

|                                                         |                      |
|---------------------------------------------------------|----------------------|
| 1870 Commissioners Publications donated free of charge: |                      |
| London PO Libraries                                     | 3                    |
| Provincial Cities and Towns UK                          | 75                   |
| Learned Centres                                         | 5                    |
| British Colonies                                        | 27                   |
| Foreign Patent offices etc                              | 29 (in 15 countries) |
| by 1882:                                                |                      |
| London and Public Offices                               | 14                   |
| Provincial Cities and Towns UK                          | 56                   |
| British Colonies                                        | 25                   |
| Foreign                                                 | 34                   |

societies and so on, the cost for a total of some 500,000 volumes of patent material had come to £503,100 by 1882. In addition to this, by 1878 a further 372 Mechanics' Institutes and other associations had been freely given the complete patent abridgements only.<sup>23</sup> Table 7 above has shown one instance of the potential impact of such information availability. Similarly, the Royal Museum and Library of Salford held full patent specifications and related literature, and the number of separate references to this local stock was reported as averaging around 5000 per annum between 1855 and 1857.<sup>24</sup> Figure 1 shows a representative batch of such material, in this case from the library of the patent office in Melbourne, Victoria for the mid-1880s.

A second tier was represented by those institutions that were generated by the inventors themselves. These included such organisations as the Inventors' Institute, of Great George Street, Westminster, which from 1862 was led astutely by Sir David Brewster, Robert Richardson CE, Henry Bessemer, C. William Siemens and Hume Williams.<sup>25</sup> Meant primarily as a lobby for further reforms and to assist inventors in reducing costs and 'to secure the advantage of certificates testifying the merits of their discoveries', the Institute also provided technical lectures, a library, laboratory and workshops.<sup>26</sup> In such professional associations as the Institution of Civil Engineers, the London Association of Foremen Engineers, and the Institution of Mechanical Engineers specialist technical lectures and discussions were common fare by the late 1850s, and such associated journals as Newton's *Journal of Arts and Science*, Peter Lund Simmonds' *The Technologist*, or William Johnson's *Practical Mechanics' Journal* provided lecture details, patent descriptions, meeting venues and so on.<sup>27</sup> In London these were conjoined with such organisations as the Polytechnic Institution, the Artisan Club, and Putney College, in addition to the better known Royal Society of Arts, as well as with the great variety of technical museums that included working machines and models.<sup>28</sup> Academic patentees such as Professor Edward Frankland of Owen's College or Professor Nicholas Callan of Maynooth extended such information systems through

#### LIST OF WORKS REFERRING TO PATENTS IN THE FREE LIBRARY OF THE PATENT OFFICE ATTACHED TO THE REGISTRAR-GENERAL'S DEPARTMENT, MELBOURNE.

*(Presented by the English Commissioners of Patents.)*

PRINTED SPECIFICATIONS OF PATENTS ENROLLED UNDER THE OLD LAW (including Surrenders and Disclaimers) from A.D. 1617 to A.D. 1852; 14,359 Specifications. 527 vols. 4to.

The same, New Law, from A.D. 1852 to A.D. 1878. About 1,775 vols. 4to.

CHRONOLOGICAL INDEXES TO PATENTS, from A.D. 1617 to A.D. 1876. 27 vols. 4to.

ALPHABETICAL INDEX, ditto ditto. 25 vols. 4to., to 1882.

SUBJECT-MATTER INDEX, ditto ditto. 27 vols., to 1880.

REFERENCE INDEX, A.D. 1617 to A.D. 1852. 2 vols. 4to.

ABSTRACTS OF SPECIFICATIONS RELATING TO SPECIAL SUBJECTS. 154 vols. 12mo.

REAPING MACHINES, Appendix to English Patents for. 1 vol. 4to.

ABSTRACTS OF REPLIES TO SECRETARY OF STATE'S CIRCULAR DESPATCHES relative to Patent Rights in the Colonies, 1853 to 1856. 2 vols. 4to.

CURTIS'S LETTERS relative to the mode of obtaining Patents in the Colonies. 1 vol. 4to.

REPORTS OF ENGLISH PATENT COMMISSIONERS, A.D. 1852 to A.D. 1866. 1 vol. 4to.

CATALOGUES OF ENGLISH PATENT OFFICE FREE LIBRARY. 7 vols. 4to.

THE COMMISSIONERS OF PATENTS JOURNAL. 38 vols. 4to.

*(Presented by the Government of New South Wales.)*

Fig. 1. Holdings of Melbourne Patent Office.



their own provincial lecturing and associational activity, this continuing a pre-reform tradition represented very well by the engineer-lecturer Henry Adcock of Birmingham.

Nor is it surprising to find that the patent-intensive urban provincial sites of Table 15 and Charts 3 and 4 above were increasingly well provided for by scientific and technical associations, and acted as a third tier that brought together engineers, tradesmen, small manufactures and professionals. Thus, from the 1830s the Sheffield Physical Club focussed on discussions and lectures on new technologies, patents, and scientific applications, all delivered by a small number of skilled tradesmen, engineers, instrument makers, industrialists and medical men. One third of known papers even prior to 1851 were on either recent patents or the science of specific industrial processes.<sup>29</sup> After 1853 this level of information dispersal was boosted by a final tier, that of government support of information and training through the Science and Art Department. Probably of considerably less importance than the three closer tiers, official financing did at least pay for the examination of science and technical classes in a multitude of provincial institutions, from mechanics' institutes to grammar and national schools, parochial schools and training colleges.<sup>30</sup>

In summary, patented invention depended upon a combination of artisan skills and engineering throughout the study period, and the trend displacement of the artisan by the engineer allowed the survival of important sites of industry-skill expertise, especially in product-based innovation. From the 1850s such inventors and innovators had at their disposal a relatively low cost system of information search and check, and at least some of the assistance available within the information system was designed to enable the artisan or craftsman to form a public company in order to exploit patented knowledge. In addition, particular urban sites, the centres of innovation, were served by a multitude of institutions created and attended by patentees. At the end of this period, urban associations were greatly aided by the emergence of professional associations and the new activities of university and college educators in the associations of the large urban areas.<sup>31</sup>

#### PATENTS AS THINGS THAT MAY BE VALUED (AS WELL AS COUNTED)

What of the more direct or technically satisfying ways of estimating the importance of patents? Here again there is much debate. So, the economist Aubrey Silberston has attempted to weigh the importance to the British economy of those industries that have been especially patent-dependent and has found on this basis that the reduction of patent protection would indeed 'seriously weaken' the economy.<sup>32</sup> Such approaches are interesting but tend to be relevant only to post-war years, insecure on the matter of measuring degrees of patent dependency (many industries are indirectly dependent on patents), and ill equipped to distinguish between the role of patents versus, say, the level of investment, the skills of labour, or the efficiency of Research and Development as such.

With the more partial data available, systematic efforts can, nevertheless be made at measuring the value of nineteenth century patents. The main case to be made is the evaluation of the *subjective commercial value of patents* by the patentees themselves. Two reasonable assumptions must be made. The first is that, however much some patentees sought aggrandisement or simple advertisement, their eye was on the main chance. That is, they would not consciously patent to their own detriment but would expect to at least obtain a reasonable rate of return upon their economic and opportunity costs, the latter embracing the time and effort taken away from other potentially lucrative efforts.<sup>33</sup> Secondly, it is



assumed that they would value their patents in exactly the same manner as any investment. So, if the market rate of return on safe investments was around 4%, the subjective valuation (market value) of an investment yielding £50 annually might be expected to be  $25 \times 50 = \text{£}1250$ . That is, in this situation an investment of £1250 at 4% would yield annually £50. Via these two assumptions this reasoning may be transferred to patents as intellectual property.<sup>34</sup> If it costs say £25 annually to renew a patent and the patentee continually renews, then he must be receiving at the very least an income from that patent asset of £25 annually, so it must be worth to him (given that he might invest that money elsewhere for the equivalent return)  $25 \times 25 = \text{£}625$  at that time.

Precisely this methodology is applied to the German patent system circa 1877–94 as shown in Table 25.<sup>35</sup> This can only be a good estimate, as patent fees are derived merely by dividing total fees from patents by the number of patents as in the third column, and then deriving the subjective capital value of the average patent in the two periods by assuming a prevailing interest rate of 4%, as in the fourth column. So if in the period 1884–89 a German businessman invested a sum of 2875 Marks, he would earn an income of 115 Marks at 4%. Subjectively, the value of a patent yielding that same income might also be calculated at 2875 Marks, assuming the inventor to be as equally rational as the businessman. From this we might calculate that in Germany the market value of the patent to contemporary patentees was on the increase precisely at the time of that nation’s rise to industrial prominence. A patent elite may also be isolated, as in Table 26.<sup>36</sup> In Germany the cost of patent re-enrolment was on an increasing sliding scale — so the first year cost 30 Marks, the tenth year 450 Marks, the final year 700 Marks. Table 26 concerns nearly 80,000 German patents divided into four re-enrolment groups. Following the established methodology it may be concluded that 65% of patents in Germany (not renewed after their fourth year) were worth approximately 2050 Marks each to their patentees, 28.2% were worth 6875 Marks, 4.7% were worth 11,875 Marks, and 1.3% were worth 16,250 Marks to each of their patentees. The elite group of patents can be regarded as comprising the last two rows, around 5% of the total or nearly 5000 patents.

TABLE 25  
Commercial valuation of German patenting, 1878–89

|         | Patent Fees<br>(P Marks) | Patent in force<br>(n) | Average Annual Fee<br>$p = P/n$ Marks | K Value patent<br>$w = 25p$ Marks |
|---------|--------------------------|------------------------|---------------------------------------|-----------------------------------|
| 1878–83 | 3566700                  | 47674                  | 74.85                                 | 1871                              |
| 1884–89 | 7976600                  | 69343                  | 115.03                                | 2875                              |

TABLE 26  
Patent longevity and patent values in Germany, 1877–94

| Period Ending<br>in Years | % Remaining<br>in Force | % in Group | Number<br>in Group | p Marks | w = 25p Marks |
|---------------------------|-------------------------|------------|--------------------|---------|---------------|
| 1–4                       | 35                      | 65         | 51422              | 82      | 2050          |
| 5–8                       | 6.8                     | 28.2       | 22309              | 275     | 6875          |
| 9–12                      | 2.1                     | 4.7        | 3718               | 475     | 11875         |
| 13–15                     | 0.8                     | 1.3        | 1004               | 650     | 16250         |

TABLE 27  
UK patent system 1853–77, applications and elite patents

|         | Applications | % sealed | % £50  | % £100  |
|---------|--------------|----------|--------|---------|
| 1853–57 | 15073        | 67.89    | 27.78  | 9.53    |
| 1858–62 | 15969        | 64.07    | 28.17  | 9.78    |
| 1863–67 | 17131        | 62.53    | 27.61  | 10.02   |
| 1868–72 | 18681        | 65.44    | 32.17  | 11.94   |
| 1873–77 | 23365        | 68.90*   | 28.79* | —*      |
| 1853–77 | 90219        | 65.84    | 29.00  | 10.39** |

\* Based on four-year average for 1873–76 and surveyed from 1879.

\*\* Based on 1853–72 only.

Table 27 applies the methodology to the 90,219 patents applied for, sealed, and maintained in Britain between 1853 and 1877. Of the £25 sealed patents, 30% paid another £75 for maintenance to a seventh year, over 10% another £125 for maintenance to the maximum fourteenth year of coverage. In this case for the elite group of 10% patents maintained to a fourteenth year, the subjective value of each patent was £3125. Calculating ambitiously from the aggregate of all group valuations the *total subjective value of all patented intellectual property in Britain 1853–1877* totals at £81,708,125. As nothing of this sort has been attempted previously, the result could be sceptically viewed. But such findings are based merely on the assumption that the rational calculation of the individual is to only spend on a patent up to the amount of the expected income return from that patent, and this is a most modest assumption for it leaves out any idea of net profit or of opportunity costs. This accords well with what is known of the £100 renewal patents in Britain, which tended to concentrate in textiles and metallurgy improvements registered by engineers or partnerships, and which included such important cases as William Siemens' 1875 renewal of the April 1868 open hearth process or Evence Coppee's renewals of his various coke oven patents from 1858. The approach yields very high values for the stock of technical knowledge captured within the patent system.

A second method of valuing patents is to focus on patenting by foreigners into other patent systems, that is, on trans-national patenting or technology transfers. Table 28 shows patent fees for nine patent systems in terms of \$US c. 1870. The right-hand column shows

TABLE 28  
Patent fees c. 1870 in \$US: nine systems

|         | A. Duration (years) | B. \$US Fee | B/A  |
|---------|---------------------|-------------|------|
| Britain | 14                  | 875         | 62.5 |
| Russia  | 10                  | 450         | 45.0 |
| Belgium | 20                  | 420         | 21.0 |
| Austria | 15                  | 350         | 23.3 |
| France  | 15                  | 300         | 20.0 |
| Italy   | 15                  | 250         | 16.7 |
| Bavaria | 15                  | 115         | 7.7  |
| Prussia | 15                  | 100         | 6.7  |
| US      | 17                  | 35          | 2.1  |

that Britain was overwhelmingly the most expensive system in which to patent from overseas, the US by far the least expensive. Yet foreign patenting into the US stood at 6% of its national total, foreign patenting into Britain at 37%, much of it from the US. This result could only be obtained in a rational world if the direct commercial return on British patenting was high or if patenting from overseas into Britain captured other gains, the most likely being entry into or participation in the buoyant information system described in the section 'Patents as Guides. . .' above.

Table 29 shows foreign competition in the British system after 1855 and is based on 3347 cases or 27% of British patenting. French and Belgian patentees fell away slightly in contrast to the rise of American and other patentees. A great deal of the work of British patent agents lay in registering, confirming, extending, challenging or defending the patent activities of such overseas patentees. A micro-analysis of such elite patent transfers is provided in British patenting into Belgium between 1860 and 1870. The shared characteristics of this group of 433 patents were that they tended to work through Brussels-based patent agents, and 75% stemmed from British partnerships — patents embraced the leading breakthroughs of this decade, including Whitworth's armaments, Neilson's steam hammer, Ramsden's steam-boilers, Fairlie's locomotive engines, and Bessemer's steel patents, but included also patents in the more competitive fields of new product technologies — saccharine, collodion and so on.

A third method of valuation is probably the most attractive and the least feasible — to estimate the 'social gain' obtaining from any one or any group of patents. Alfred Marshall once generalised that the greatest notion of the industrial revolution was 'the great architectonic principle that a well driven machine tool could become the parent of new machine work more exact than itself, which could become in its turn the parent of yet more exact machines, and so on.' Because of such generational impacts and other effects, the private gain of the inventor 'is seldom a hundredth part — sometimes not a millionth part — of the *social gain*.'<sup>37</sup>

One possibility is to investigate special pleas for patent extension in such legal sites as the Privy Council in Britain or the Commissioner of Patents in the United States. By the 1860s an informal rule by precedent seems to have related to the amount of the patentee's profit as a ratio of the social or public return to the patent. As Brougham argued before the Privy Council, even if there had been an excellent private return, a case could be made out for extended or broadened protection of the patentees if the ratio of private to public gain remained very small. That contemporaries thought they could make such estimates is shown in the case of US sewing machines, where the legal case for the patentee was based on a New York figure that showed the ratio for that state alone to stand at 1:103, at which time sewing machines in the US numbered some 100,000. On a similar basis Goodyear showed

TABLE 29  
Percentages of all direct foreign patentees 1855–70

|               | 1855  | 1860  | 1865  | 1870  |
|---------------|-------|-------|-------|-------|
| France        | 63.41 | 65.89 | 53.15 | 48.47 |
| Belgium       | 4.88  | 8.45  | 4.33  | 3.57  |
| USA           | 17.89 | 14.87 | 31.10 | 32.14 |
| Other Foreign | 13.82 | 10.79 | 11.42 | 15.82 |

a ratio for the India rubber patent of 1:40, arguing that if ‘this process is worth two millions of dollars, the applicant has received but a little more than one-fortieth part of the remuneration which he was entitled to claim’, so that the company deserved its patent extension.

### CONCLUSIONS — ON COUNTING SMALL CHANGE

The great variety of patent data sampled in this paper gives some notion of its potential value in historical analysis. Patent trends appear to illustrate broad patterns of industrial development, as well as such more specific events as the American Civil War or the German technical invasion of the British coal-tar industry and market. Patterns within nations show something about the social-geography of inventive activities. Specifically the shift of inventive activity to the industrial regions and to large urban sites can be measured, and it may be seen that some industrial cities were more intensive incubators of activity than others. A shift from artisan invention towards the engineer undoubtedly took place from the 1850s, and this might be seen as evidence confirming the Rosenberg approach. On the other hand artisan invention continued to thrive in urban provinces, especially in product innovation, and this appears to have been facilitated by partnerships and by an urban culture of scientific and technical association. Within such sites of inventive activity at a time of few formal qualifications or marks of esteem, patent registrations and extensions may well have acted as markers within a dynamic information system. It also seems quite possible that we might hazard estimates of the value of intellectual property as defined by national patent systems. There are several methods of approaching this, and that of subjective commercial valuations based on the patentee as rational man has been suggested. Alongside partnerships, legal challenges and transnational registrations, such valuations may also help in the general isolation of elite patents from the general mass. As emphasised at the outset, not all patents were equal, and counting small change may turn out to be a surprisingly hazardous venture.

### NOTES AND REFERENCES

1. Unless otherwise stated or obvious, the data for this paper is extracted from the original patent applications in the British patent system, each of which for cited years has been sighted and summarised on an EXCEL spreadsheet system. All other data unless mentioned otherwise is derived from systematic data in *The Commissioner of Patents Journal*, annually from 1852 to 1883. For more detail of these and other sources see below Inkster 1990 and 2000. The basic research was originally supported by two research grants from NTU.
2. Examples of work of potential interest to engineering historians include F.D. Prager, ‘A history of intellectual property from 1545 to 1787’, *Journal of the Patent Office Society*, vol. 26 (1944), pp. 711–60; A.R. Pred, *Urban Growth and the Circulation of Information. The United States System of Cities 1790–1840* (Harvard University Press, 1973); Harry Dutton, *The Patent System and Inventive Activity during the Industrial Revolution* (Manchester University Press, Manchester, 1984); Christine MacLeod, *Inventing the Industrial Revolution. The English Patent System, 1660–1800* (Cambridge University Press, Cambridge, 1988); Pamela O. Long, ‘Invention, authorship, “intellectual property” and the origin of patents’, *Technology and Culture*, vol. 32 (1991), pp. 846–84; *Inventions, innovations et espionnage du savoir technique dans l’Europe moderne*, eds Marco Belfanti and Fabio Giusberti, special issue, *History and Technology*, XVI, 3 (2000); Liliane

- Hilaire Perez, *L'invention technique au siècle des Lumières* (Albin Michel, Paris, 2000). Work by Ian Inkster on patents has also embraced the themes of technology transfer, relations between artisanal and engineering patenting, and investigations of the intellectual milieu of patentees, and this includes 'The Ambivalent Role of Patents in Technology Development', *Bulletin of Science, Technology and Society*, vol. 2 (1982), pp. 181–90; 'Intellectual Dependency and the Sources of Invention. Britain and the Australian Technological System in the Nineteenth Century', *History of Technology*, vol. 12 (1990), pp. 40–64; 'Technology Transfer in the Great Climacteric: Machinofacture and International Patenting in World Development circa 1850–1914', *History of Technology*, vol. 21 (1999), pp. 87–106; 'Machinofacture and Technical Change: The Patent Evidence' in Ian Inkster et al., *The Golden Age. Essays in English Social and Economic History, 1850–1870* (Ashgate, Aldershot, 2000), pp. 121–39; and 'Artisans de la découverte. Modèles Britanniques et Internationaux d'Innovation Technologique 1700–1914', *International Symposium Artszans industrie. Nouvelles révolutions* (Paris, France, June 2001).
3. Eric Hopkins, *The Rise of the Manufacturing Town. Birmingham and the Industrial Revolution* (Sutton Publishing, Stroud, 1998), chapters 2–4. For the inventive context see R.B. Prosser, *Birmingham Inventors and Inventions* (Birmingham, 1881); Maxine Berg, 'Commerce and creativity in eighteenth century Birmingham', in *Markets and Manufacturers in Eighteenth Century England*, ed. Berg (Routledge, London, 1991), pp. 173–204.
  4. Both highly dependent technological systems in areas of recent white settlement with either strong commercial or colonial relations with one overwhelming industrial power — see Inkster, *Technology and Industrialisation* (Ashgate, London, 1998), pp. 5–8 (1) and essays IV–VI. Patents do not at all reflect rankings of nations in terms of GDP per head — for instance, in such terms in 1870 the top ranked nations were Australia, UK, Belgium, Netherlands and Switzerland. So in the study period patenting was not a function of wealth per se but rather of level of manufacturing industrialisation, which is what would be predicted on the assumption that patenting was a significant technological activity and measurement.
  5. *Report of the Commissioners on the Working of the Laws relating to Letters Patent for Inventions, Both Houses* (London, House of Commons, 1865), p. 152.
  6. Symptomatically, Caro had previously been employed working on aniline dyes as chemist and colourist to Roberts, Dale and Co. in Manchester: see E. Homburg, 'The emergence of research laboratories in the dyestuffs industry, 1870–1900', *The British Journal for the History of Science*, vol. 25 (1992), pp. 91–111; C. Reinhardt, 'An instrument of corporate strategy: The central research laboratory at BASF 1868–1890', in *The Chemical Industry in Europe 1850–1914*, eds Homburg, A.S. Travis and H.G. Schroter (London, Dordrecht, 1998), pp. 239–59.
  7. 'The Coal Tar Colour Industry in Germany and England', *Nature* (12 December 1901), pp. 138–39; L.F. Haber, *The Chemical Industry during the Nineteenth Century* (Oxford, 1958), especially p. 130 passim.
  8. It is worth noting that two authoritative historians in this field have generalised on the basis of their own and others' research that the new corporate research laboratories in fact 'preserved strong elements of testing, quality control, and investigations directed at the protection of patents, rather than a preoccupation with the development of new products and processes, a stress on tactical commercial-mindedness that distinguished them from universities, polytechnics and government institutions' — see R. Fox and A. Guagnini, *Laboratories, workshops and sites. Concepts and practices of research in industrial Europe, 1800–1914* (Office for History of Science and Technology, University of California, Berkeley, 1999), quote p. 149.
  9. The other British firms were Claus and Ree (90), Brooke, Simpson and Spiller (7), and W.G. Thompson (2): see Haber, *ibid.*, pp. 126f, and F. Rose, *Chemical Instruction in Germany and the Growth and Present Conditions of the German Chemical Industry* (Diplomatic and Consular Reports, London, 1901).
  10. *Japan Year Book*, various years 1904–16; *Patent and Trade Mark Review* various years 1912–22; *Bureau des Brevets d'Invention, Lois concernant la protection de la propriété industrielle dans L'Empire du Japon* (exposition Universale de 1900 Paris, Paris, 1900–01).

11. Ian Inkster, *Japanese Industrialisation. Historical and cultural perspectives* (Routledge, London, 2001), especially chapters 3 and 4.
12. Simon Sharma, *The Embarrassment of Riches. An Interpretation of Dutch Culture in the Golden Age* (Collins, London, 1988); *Urban Achievement in Early Modern Europe. Golden Ages in Antwerp, Amsterdam and London*, eds Patrick O. Brien et al. (Cambridge University Press, Cambridge, 2001).
13. For discussion of public policy attempts at the artificial creation of innovative urban sites see especially Sheridan Tatsuno, *The Technopolis Strategy* (Prentice Hall, New York, 1986); T. Morris Suzuki, *Beyond Computopia* (Kegan Paul International, London, 1988); Ian Inkster, *Clever City* (Sydney University Press and Oxford University Press, Melbourne, 1991).
14. It must be acknowledged that a prime objective is to differentiate more general engineering skills and training from the industrial skills of trade apprentices, which we may designate approximately 'artisan'. For this period, the Engineer category includes all self-designated engineers, mechanics, wrights and millwrights, chemists and foremen. Artisan embraces all patentees who define themselves in terms of their trade apprenticeship, other than the engineers.
15. For the historical details of which see *Metropolis and Province. Science in British Culture 1780–1850*, eds Inkster and Jack Morrell (University of Pennsylvania Press, Philadelphia, 1983), for instance chapters 1, 5, 7 and 8, and Ian Inkster, *Scientific Culture and Urbanisation in Industrialising Britain* (Ashgate 1997), especially chapters 4, 5, 9, 10 and 11.
16. For Britain in particular see James B. Jefferys, *The Story of the Engineers* (Lawrence and Wishart, London, 1945); Charles More, *Skill and the English Working Class 1870–1914* (Croom Helm, London, 1980); and R.A. Buchanan, *The Engineers: A History of the Engineering Profession in Britain 1750–1914* (London, 1989).
17. Nathan Rosenberg, *Perspectives on Technology* (Cambridge University Press, Cambridge, 1975); idem., *Inside the Black Box: Technology and Economics* (Cambridge University Press, Cambridge, 1982).
18. Ubiquitous because widely spread through self-serving self-reference! The label of course captured many merchants and some large manufacturers, and as the patent system became more formalised the category shrunk. Local research also serves to reduce this label to its true component parts.
19. William Spence, *Patentable Invention and Scientific Evidence* (London, Stevens and Norton, 1851), pp. x–xii, 14–17.
20. William Spence, *A Treatise on Principles Relating to the Specification of a Patent for Invention* (V. and R. Stevens and G.S. Norton, London, 1847), p. 73.
21. *Newton's London Journal of Arts and Science*, X (1859), pp. 129–34, 257–63.
22. French inventors used the services of the leading London patent agents in profusion, flooding the portfolios of agents such as W.R. Lake of the International Patent Office, Southampton Buildings, who through his other firm Haseltine, Lake and Co. was the largest in the country during the 1870s, A.M. Clark of Chancery Lane, and H.A. Bonneville of the British and Foreign Patent Office, Piccadilly.
23. *Commissioner of Patents Journal*, II, (July–December, 1878), pp. 990–92.
24. *Report of the Commissioners on the Working of the Laws Relating to Letters Patent for Inventions, Both Houses of Parliament*, (House of Commons, London, 1865), p. 5.
25. *Report of the Council of the Inventors' Institute* (London, 1863). This institute was designed to act as a counterweight to Ricardian theory, which tended towards an argument that free markets were best in all possible contexts, and that patents represented a restriction on the free market for technology and technological information. In fact, patents helped to define that market.
26. The Inventors Institute, *Report of the Council of the Institute made to the First Annual Meeting, 14 May 1863* (London, 1863), quote p. 7.
27. Small groups such as the London Association of Foremen Engineers (founded in 1852) were designed as both benefit societies and forums for practical information, and were normally led by



- successful patentees or engineers — in this case by the ever-active Newton and by W. Keytes, manager of the Royal Gun Factories at Woolwich.
28. In London by the early 1860s these included the South Kensington Museum, Museum of Practical Geology (Jermyn Street), Museum of Economic Botany (Kew Gardens), Industrial Museum (Crystal Palace), Museum of the Pharmaceutical Society (Bloomsbury Square), East India House Museum (Leadenhall Street), Economic Museum (Perryn House, Twickenham), and the Museum of the Botanical Society (Regents Park).
  29. *Sheffield Mercury*, 21 June, 16 August 1834; 21 October 1837.
  30. *Report from the Select Committee on Scientific Instruction together with the Proceedings of the Committee, Minutes of Evidence* (House of Commons, London, 1868), in addition, the Schools of Design had been established in 1837 under recommendation of a select committee of the House of Commons in 1835, but were put under the Department of Practical Art from 1852.
  31. Ian Inkster, 'Association, provincialisme et sociologie du progrès technique: le cas de Grande-Bretagne, c. 1780–1914', Colloque Historique, Société d'encouragement pour l'Industrie Nationale, Paris, France, 8–9 November 2001.
  32. Aubrey Silberson, 'The Economic Importance of Patents', *Technology and Economic Progress*, ed. in Silberson (Macmillan, London, 1989), pp. 155–80, quote p. 178.
  33. This is the major assumption, from which all else follows by calculation. If the assumption is to be seriously disputed, the critic must go beyond a bit of anecdotal evidence as to frivolous inventors or inventions. It might be remembered that at a time of no or contentious qualifications, mechanics, engineers and small producers had to make a name in some way, a name that was worth a commercial value. If in some instances they did this through patenting of a form that we now see as trivial, this does not remove the case. Again, trivial patents in the technical sense (that is, involving a minor alteration, such as a small change in the strength of a chemical solution) might have been of real value in reducing the cost of applications, removing a monopoly of supply, or allowing production with different or 'inferior' quality of raw material inputs — an excellent example is the patent incrementalism attached to the cyanide process for gold production in the last years of the century, for which in summary see Ian Inkster, *Technology and Industrialisation* (Ashgate, London, 1998), pp. 5–8 (I), and for the details, Jan Todd, *Colonial Technology. Science and the Transfer of Innovation to Australia* (Cambridge University Press, Cambridge, 1995).
  34. For an example of such reasoning in the contemporary engineering literature see that of Dr J. Ephraim, nos 36 and 37 for 1901 of *Zeitschrift für angewandte Chemie*.
  35. *Chemische Zeitschrift*, 14 May 1890, calculations from data at p. 303.
  36. *Dingler's Polytechnisches Journal* (1894), calculated from tables and data pp. 160–64.
  37. Alfred Marshall, *Industry and Trade, A Study of Industrial Technique and Business Organisation* (Macmillan, London, 1919), quote pp. 206, 325.